Exercises: DEEP LEARNING

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1. For $y = (2x+1)^2$, the derivative is:

A. $4(2x+1)$ A&B
B. $2(2x+1) \cdot 2$
C. $8x+2$
D. All of the above

2. For $y = (5x-3)^3$, the derivative is:

A. $3(5x-3)^2$
B. $15(5x-3)^2$
C. $(5x-3)^2$
D. $5(5x-3)^3$ B

3. For $y = \sqrt{3x+1}$, the derivative is:

A. $\dfrac{1}{2\sqrt{3x+1}}$
B. $\dfrac{3}{2\sqrt{3x+1}}$
C. $\dfrac{3}{\sqrt{3x+1}}$
D. $\sqrt{3x+1}$ B

4. For $y = \sin(4x)$, the derivative is:

A. $\cos(x)$
B. $4\sin(4x)$
C. $4\cos(4x)$
D. $-4\sin(4x)$ C

5. For $y = e^{2x^2}$, the derivative is:

A. e^{2x^2}
B. $4xe^{2x^2}$
C. $2xe^{2x}$
D. $4x^2e^{2x^2}$ B

6. For $z = (xy)^2$, the partial derivative $\dfrac{\partial z}{\partial x}$ is:

A. $2xy$
B. $2y(xy)$ B & C
C. $2xy^2$
D. y^2

7. For $u = x+y$ and $z = u^3$, $\dfrac{\partial z}{\partial y}$ is:

A. $3u^2$
B. $3(x+y)^2$
C. Both A and B
D. $3(x+y)$ C

8. For $f(x,y) = x^2+y^2$, gradient at $(3,4)$ is:

A. $(6,8)$ A
B. $(3,4)$
C. $(9,16)$
D. $(2,2)$

9. A single gradient descent update is closest to:

A. Random search
B. Taking one step downhill B
C. Jumping to the minimum
D. Averaging data

10. In a neuron $h = wx+b$, the derivative $\dfrac{\partial h}{\partial w}$ is:

A. x A
B. 1
C. h
D. w

11. For output $y = h^2$, the gradient $\dfrac{\partial y}{\partial h}$ is:

A. h
B. $2h$
C. $1/h$ B
D. h^2

12. Backpropagation requires repeated use of:

A. Chain rule A
B. Quotient rule
C. Product rule only
D. Logarithms

13. In gradient descent, weights update according to:

A. Loss direction D
B. Gradient magnitude
C. Learning rate
D. All of the above

14. The gradient of the bias b in a neuron is always:

A. x
B. 1 B
C. b
D. Varies randomly

15. Gradient descent reduces loss because:

A. It finds the steepest upward direction
B. It finds the steepest downward direction
C. It jumps to the minimum B

D. It memorizes data

16. For $f(w) = 3w^2 + 2w$, the gradient is:

A. $6w + 2$
B. $3w^2$ A
C. $3(2w)$
D. $w + 2$

17. The gradient of a constant is:

A. 0 A
B. 1
C. Undefined
D. Same as the constant

18. The derivative of $y = \ln(5x)$ is:

A. $1/x$
B. $5/x$
C. $5\ln(x)$ A
D. $1/(5x)$

19. For $f(x) = x^3$, the gradient at $x = 0$ is:

A. 3
B. 0 B
C. 1
D. Does not exist

20. The gradient of $f(x) = x^{-1}$ is:

A. $-x^{-2}$ A
B. x^{-2}
C. $-x$
D. $1/x^2$